SVD proof continued:

Form $V = [v_1 | v_2 | \cdots | v_m] \in M_m(\mathbb{R})$

– also an orthogonal matrix.

Take $\Sigma = \begin{bmatrix} \sigma_1 & & & & & \\ & \sigma_2 & & & & \\ & & \ddots & & & \\ & & & \sigma_r & & \\ & & & & 0 & \\ & & & & & \ddots \end{bmatrix} \in M_{n,m}(\mathbb{R})$

where $\sigma_j = \sqrt{\lambda_j} \in \mathbb{R}_+$, $j = 1, 2, \ldots, r$.

Then $AV = [Av_1 | Av_2 | \cdots | Av_r]$

$$\overset{\text{defn}}{\underset{\text{of } u_j}{=}} [\sigma_1 u_1 | \sigma_2 u_2 | \cdots | \sigma_r u_r | 0 | \cdots | 0] \quad (*)$$

$$= U\Sigma$$

$$\Rightarrow \quad A = U\Sigma V^{-1} = U\Sigma V^T$$

(Note: $(*)$ $u_j = \dfrac{Av_j}{\|Av_j\|_2} = \dfrac{Av_j}{\underbrace{\|v_j\|}_{=1}\sqrt{\lambda_j}} \Rightarrow$

$Av_j = \sqrt{\lambda_j}\, u_j = \sigma_j u_j$)

∎

Two equivalent formulations of SVD

(1) Let $A \in M_{n,m}(\mathbb{R})$, $r = \operatorname{rank} A > 0$.

Then $\exists$ orthonormal bases

$$u_1, \dots, u_n \in \mathbb{R}^n$$
$$v_1, \dots, v_m \in \mathbb{R}^m$$

and $\exists\ \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$

such that

$$A v_i = \begin{cases} \sigma_i u_i, & i = 1, 2, \dots, r \\ 0 & i = r+1, \dots, m \end{cases}$$

& $$A^T u_i = \begin{cases} \sigma_i v_i, & i = 1, 2, \dots, r \\ 0 & i = r+1, \dots, m \end{cases}$$

(2) $A = U \Sigma V^T$ (SVD) can be written as

(#) $$\boxed{A = \sum_{j=1}^{r} \sigma_j u_j v_j^T}$$ (sum of r rank-one matrices)

This can be viewed as a generalization of the spectral theorem, where $A = A^T$

$$\left(A = A^T \implies A = \sum_{j=1}^{n} \lambda_j u_j u_j^T \right)$$

how is (#) used in practice

applications in biology
dmaging data

Mystery: In "real life", matrices have some singular values that are significantly bigger that the rest

So the following "economization" is possible.

Say $\sigma_1 \geq \sigma_2 \ldots \geq \sigma_k >> \sigma_{k+1} \geq \sigma_{k+2} \cdots \sigma_r > 0$

Then

$$A = \sum_{j=1}^{r} \sigma_j u_j v_j^T \approx \sum_{j=1}^{k} \sigma_j u_j v_j^T$$

quantify if possible approximation works well in practice.

(I'll try to show it on Wednesday)

Some of the reason it works well will be evident next:

SVD is connected to a lot of concepts we've seen:

<u>Theorem</u> 4.2.1 Let $A \in M_n(\mathbb{R})$. Then

$$\|A\|_2 := \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = \sigma_1$$

<u>Proof</u> Recall $Av_1 = \sigma_1 u_1$ (in the proof of SVD). So

$$\|A\|_2 \geq \frac{\|Av_1\|_2}{\|v_1\|_2} = \frac{\|\sigma_1 u_1\|_2}{\|v_1\|_2}$$

$$= \sigma_1 \frac{\overset{=1}{\cancel{\|u_1\|_2}}}{\underset{=1}{\cancel{\|v_1\|_2}}} = \sigma_1$$

So $\boxed{\|A\|_2 \geq \sigma_1}$ (1)

Let $x \in \mathbb{R}^n$. Write $x = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$

Thus $\|x\|_2^2 = |c_1|^2 + |c_2|^2 + \cdots + |c_n|^2$

Also $Ax = c_1 Av_1 + \cdots + c_n Av_n$

$= c_1 \sigma_1 u_1 + \cdots + c_n \sigma_n u_n$

$$\|Ax\|_2^2 = |\sigma_1 c_1|^2 + \cdots + |\sigma_n c_n|^2$$

$$\leq \sigma_1^2 \left(|c_1|^2 + \cdots + |c_n|^2 \right)$$

$$= \sigma_1^2 \|x\|_2^2$$

So $$\frac{\|Ax\|_2^2}{\|x\|_2^2} \leq \sigma_1^2$$

i.e. $$\boxed{\frac{\|Ax\|_2}{\|x\|_2} \leq \sigma_1} \quad (2) \quad (\Rightarrow \|A\|_2 \leq \sigma_1)$$

By (1) & (2)

$$\boxed{\|A\|_2 = \sigma_1}$$

∎

10/23/24

<u>Theorem 4.2.4</u> Let $A \in M_n(\mathbb{R})$ be nonsingular. Suppose $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n > 0$ are the singular values of A. Then

$$\kappa_2(A) = \|A\|_2 \|A^{-1}\|_2 = \frac{\sigma_1}{\sigma_n}.$$

<u>Proof</u> As usual, let $A = U\Sigma V^T$ be the SVD of A ($\Sigma = \operatorname{diag}(\sigma_1, \sigma_2, \dots, \sigma_n)$)

Then

$$A^{-1} = (V^T)^{-1}\Sigma^{-1}U^{-1}$$
$$= (V^{-1})^T\Sigma^{-1}U^T = (V^T)^T\Sigma^{-1}U^T$$
$$= V\Sigma^{-1}U^T,$$

which is a SVD of A^{-1}; also

$$\Sigma^{-1} = \begin{pmatrix} 1/\sigma_1 & & & \\ & 1/\sigma_2 & & \\ & & \ddots & \\ & & & 1/\sigma_n \end{pmatrix},$$

except that the singular values are ordered in decreasing order:

$$1/\sigma_n \geq \frac{1}{\sigma_{n-1}} \geq \dots \geq \frac{1}{\sigma_1} > 0$$

So the singular values of A^{-1} are

$$\frac{1}{\sigma_n} \geq \frac{1}{\sigma_{n-1}} \geq \dots \geq \frac{1}{\sigma_1} > 0.$$

Thus, by the previous theorem,

$$K_2(A) = \|A\|_2 \|A^{-1}\| = \sigma_1 \cdot \frac{1}{\sigma_n} \quad \blacksquare$$

More SVD considerations

- **Numerical rank determination**

  Find rank of $A \in M_{n,m}(\mathbb{R})$.

  Finding Echelon form is inefficient.

  Use SVD

  Find SVD and decide which is the minimal singular value σ_r. Then

  $$r = \text{rank}(A) = \#\text{ of nonzero singular values}$$

  A decision is as to what is "0"?

This can be problematic too:

<u>Example</u> $A = \begin{bmatrix} 1/3 & 1/3 & 2/3 \\ 2/3 & 2/3 & 4/3 \\ 1/3 & 2/3 & 3/3 \\ 2/5 & 2/5 & 4/5 \\ 3/5 & 1/5 & 4/4 \end{bmatrix}$

$\sigma_1 = 2.5987, \quad \sigma_2 = 0.3682, \quad \sigma_3 = 8.6614 \times 10^{-17}$

"Declare" (decide), that $\sigma_3 \approx 0$, so rank $A = 2$. Check other ways ....

You'll find out rank $A = 3$.

<u>Practice</u> If $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k \gg \varepsilon \geq \sigma_{k+1} \geq \cdots \geq \cdots$

and $\varepsilon = 10 \cdot u \cdot \|A\|_2 = 10 \cdot u \cdot \sigma_1$

u = roundoff decimal limit.

then we declare rank $A = k$.

Moore-Penrose inverse (aka pseudoinverse) of a matrix A

Definition Let $A \in M_{n,m}(\mathbb{R})$. Then there exists unique $B \in M_{m,n}(\mathbb{R})$ such that

(1) $ABA = A$

(2) $BAB = B$

(3) $(AB)^T = AB$

(4) $(BA)^T = BA$

We call B the Moore-Penrose inverse of A, and denote it by $B = A^+$

Notes

(i) A^+ always exists & unique

(ii) Existence is justified via SVD

(iii) If $m = n$ & A^{-1} exists, then $A^+ = A^{-1}$.

How to find A^+

- $A \in M_{n,m}(\mathbb{R})$
- Compute SVD : $\boxed{A = U \Sigma V^T}$

$$\Sigma = \mathrm{diag}(\sigma_1, \sigma_2, \dots, \sigma_r, 0, \dots, 0)$$
$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$$

- Define

$$\Sigma^+ = \mathrm{diag}\left(\frac{1}{\sigma_1}, \frac{1}{\sigma_2}, \dots, \frac{1}{\sigma_r}, 0, \dots, 0\right)$$

- Then

$$\boxed{A^+ = V \Sigma^+ U^T}$$

<u>Proof</u> Verify the 4-properties:

(1) $$AA^+A = U\Sigma \underbrace{\cancel{V^T V}}_{I} \Sigma^+ \underbrace{\cancel{U^T U}}_{I} \Sigma V^T$$
$$= U \Sigma \Sigma^+ \Sigma V^T = U \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & 0 & \\ & & & & \ddots \; 0 \end{bmatrix} \Sigma V^T$$
$$= U \Sigma V^T = A$$

(2) - (4) Similar. ∎

▶ Connection to LSP (next time)

<u>Note</u> SVD & everything else generalize easily to $A \in M_{n,m}(\mathbb{C})$

$$A = U \Sigma V^*$$

$\Sigma = \operatorname{diag}(\sigma_1, \dots, \sigma_r, 0, \dots 0)$

σ_j^2 eigenvalues of A^*A

Columns of U are eigenvectors of A^*A, Column of V are eigenvectors of AA^*

$$\left.\begin{aligned} U^* &= U^{-1} \\ V^* &= V^{-1} \end{aligned}\right\} \text{ unitary matrices}$$

($*$ means "conjugate transpose")

(Announcements on Friday)

10/25/24

Connection of A^+ to the LSP $Ax = b$

Theorem If $Ax = b$, then $x = A^+ b$ is a least-squares solution of minimal length: That is,

(1) $\| A(A^+ b) - b \|_2$ is minimal among $\|Ax - b\|_2$ for all x

(2) $\| A^+ b \|_2 \leq \| \hat{x} \|$ for all least-squares solutions $\hat{x}$.

Proof

Case (i): $Ax = b$ is consistent, i.e. $\exists x_0 : Ax_0 = b$

Then
$$AA^+Ax_0 = Ax_0 = b$$
$$\Rightarrow A(A^+ b) = b$$
$$\Rightarrow A^+ b \text{ is a solution.}$$

In this case A^+b is a unique minimal solution:

Every solution to $Ax=b$ looks like

$$z = A^+b + w, \quad w \in N(A)$$

But $A^+b \in R(A^+) \overset{*}{=} R(A^T) \overset{\text{FTLA II}}{=} N(A)^\perp$

Thus
$$\|z\|_2^2 = \|A^+b + w\|_2^2$$
$$\overset{\text{P.T.}}{=} \|A^+b\|_2^2 + \|w\|_2^2$$
$$\geq \|A^+b\|_2^2$$

$$\Rightarrow \quad \|z\|_2 \geq \|A^+b\|_2$$

which proves A^+b is a minimal solution and thus unique.

Case (ii): $Ax=b$ is inconsistent.

Then $A^TAx_0 = A^Tb$ is consistent & the solution x_0 are least-squares solutions.

(Normal equations)

But $x_0 = A^+ b$ satisfies the normal equations: Let $A = U\Sigma V^T$ be the SVD.

Then $$(A^TA)(A^+b) = V\Sigma^T \underbrace{U^TU}\Sigma V^T(A^+b)$$

$$\underset{\substack{\uparrow \\ \text{previous thm}}}{=} V\Sigma^T\Sigma \underbrace{V^T(V}\Sigma^+U^T)$$

$$= V\Sigma^T \underbrace{\Sigma\Sigma^+}_{= \text{ Since } \Sigma \text{ diagonal}} U^Tb$$

$$= V\Sigma^TU^Tb = A^Tb.$$

Thus A^+b is a least-squares solution. Minimality & uniqueness follow exactly as in case (i) ∎

<u>Note</u> (*) $R(A^+) = R(A^T)$ is left as an exercise. (Hint: show $\subseteq$, $\supseteq$)

Announcements:

(1) Monday: Start Chapter 5
Eigenvalues

(2) Wednesday: No class
Midterm will show
up on Canvas.